Jose Sanchez Gonzalez

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SUMMARY

M.S. Artificial Intelligence graduate measuring model inference performance and numerical agreement. Python/C++ and PyTorch; V100 precision benchmarks, ONNX parity checks, and statistically evaluated GPU reranking.

EDUCATION

Oregon State University Corvallis, OR
Master of Science in Artificial Intelligence Jun 2026
Bachelor of Science in Computer Science (Applied: Data Science); Mathematics Minor Jun 2025

TECHNICAL SKILLS

Programming and ML: Python, C++17, PyTorch, NumPy, Linux
Inference: fp16/fp32 benchmarking, ONNX export/parity, Docker, FastAPI
Evaluation: Paired comparisons, matched controls, pytest, gtest, GitHub Actions
Integration: PostgreSQL/pgvector, ROS 2 replay, Isaac Lab, MuJoCo

EXPERIENCE

Graduate Teaching Assistant | Oregon State University Sep 2025 – Jun 2026

- Led weekly two-hour studios for 25+ students in Operating Systems and Computational Methods, teaching C++ concurrency, memory management, file systems, debugging, and numerical simulation workflows.
- Coordinated with course staff to diagnose recurring technical problems, designing targeted demonstrations and communicating debugging strategies across laboratory sessions.

Math and Computer Science Tutor | Oregon State University Nov 2022 – Jun 2025

- Mentored 50+ students one-on-one across mathematics, algorithms, and programming, identifying knowledge gaps and coaching structured quantitative problem solving and independent software debugging.

PROJECTS

SPUR: Inference Performance and Numerical Validation | PyTorch, ONNX, Docker, FastAPI

- Benchmarked six-view PyTorch refiner-only inference at 156 ms p50 in fp16 versus 393.5 ms in fp32 on a V100 (200 calls; excludes depth preprocessing and HTTP).
- Validated separate split-ONNX export against PyTorch (max absolute error 1.53e-5); packaged Docker/FastAPI inference.
- Corrected camera intrinsics across four views of one synthetic tree/shot, reducing median predicted-depth surface error from 69.5 mm to 7.4 mm.

MetaNaviT: GPU Reranking and Retrieval Evaluation | Python, BGE, PostgreSQL, pgvector, BM25

- Optimized GPU reranking from 923 to 191 ms p50 with fp16 and a 512-token cap; quality remained statistically tied on 300 SciFact queries.
- Measured +0.075 paired nDCG@10 over BM25 on 300 queries (95% CI 0.053–0.101); generated evidence tables with result-drift checks.
- Contributed retrieval/data APIs on a six-person capstone team; independently extended evaluation, fixed silent reranker fallback, and checked backend parity on 136 queries.

Berkeley Humanoid Lite: Arm Calibration and Policy Evaluation | Quest 2, SocketCAN, Isaac Lab, ONNX

- Customized the robot with a pan/tilt stereo camera, lidar, and an IMU; built AR features and sensor/calibration diagnostics for teleoperating the arms from Quest 2 through inverse kinematics and SocketCAN.
- Calibrated all ten physical arm joints, setting per-joint torque limits of 1.25–5.25 Nm; one wrist settled within 0.86 degrees on +/-10-degree step tests.
- Organized a five-setting, 450-episode flat-ground comparison with matched controls and rollout artifacts; exported Isaac Lab policies to ONNX for headless MuJoCo evaluation.
- Trained planner-free PPO navigation reaching 35/36 goals across three seeds on the same twelve unseen mazes, with no falls and oracle pose/goal in simulation.

Pruning: C++ Geometry and Software Replay | C++17, Python, ROS 2, gtest

- Ported 8x8 time-of-flight deprojection to C++17 with gtest; matched Python reference geometry on 232 recorded zones at 1e-9 m tolerance.
- Reproduced 199/199 recorded decision states in ROS 2 software replay; RGB-blackout and depth-dropout controls each withheld approach and release on 39/39 frames.